EE1007
Introduction to Computer Science Laboratory

Lab 4: Control Statement II (Repeat Control)

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P110009: Lab 4.1 Prime Factorization

- In this program, you need to write a program that can decompose the prime factors of any positive integer
- Then, the prime factors are displayed in increasing order
- The program need to design fool-proof to avoid program errors
- Please use **if… else**, **while** or **do… while** to implement your program, otherwise you will obtain 0 point
- The range of each integer is $[2 \sim 2^{31}-1]$

P110009: Lab 4.1
Prime Factorization

◆The screen dialog should appear as follows:

```
Enter a number: -222
Please enter a correct number:1
Please enter a correct number:1260
1260 = 2*2*3*3*5*7

Process returned 0 (0x0)   execution time : 11.066 s
Press any key to continue.
```


P110010: Lab 4.2
Weekly calculation

- ◆ In this problem, you need to write a program which can find the corresponding day of a given date
- ◆ 2021.01.01 is Friday
- ◆ The program need to design fool-proof to avoid program errors
- ◆ Please use **switch…case** to complete this program, **do not use if…else** otherwise no score will be given.
- ◆ The range of each integer is [20210101~2^{63}-1].

Lab 4.2
Weekly calculation

- In this program, you **cannot** use any formulas to calculate the day.
- For example,
  - 基姆拉爾森計算公式
    W = (d + 2 * m + 3 * (m + 1) / 5 + y + y / 4 - y / 100 + y / 400) mod 7
  - 蔡勒公式
    W = [C/4] - 2C + y + [y/4] + [13 * (M+1) / 5] + d - 1

Lab 4.2
Weekly calculation

- The screen dialog should appear as follows:

```
Enter a date (ex. 20211014): 20200101
This date does not exist or is within the calculation range. Please enter a correct date (ex. 20211014): 20223113
This date does not exist or is within the calculation range. Please enter a correct date (ex. 20211014): 20211014
20211014 is Thursday.

Process returned 0 (0x0)   execution time : 32.933 s
Press any key to continue.
```

Evaluation

- Lab4.1(40%)
  - 5 testcases, 8% each
- Lab4.2(60%)
  - 5 testcases, 12% each





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